

CANALRUNG · REQUIREMENTS

CanalRung requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions of 2026-10-03 (4A and 9A, CNR-DDR-003): R2 mass target restated to 4.5 kg; R6 restated for the 500 x 1,200 mm footing as the standard at every site, in drained or saturated soil; R4 and R9 results updated for the 9 mm ropes and larger footing
0.3	2026-10-03	Amish Chadha	TRL 3: status of every requirement from CNR-CAL-001 on the constructable design (CNR-DDR-002); R2 not met on mass; R9 reported against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2: measurable targets; R1 restated for a stated design envelope; R4 and R6 given margins; R6 restated for the fixed station anchor; R10 added (nothing thrown can hurt the person in the water); decided under Amish's 2026-10-03 pre-approval (CNR-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

CanalRung requirements

CanalRung meets nine of its ten requirements on paper or by design at TRL 3 and is over its value-engineering target on the tenth (R9). On 2026-10-03 Amish decided "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A": for CanalRung, 4A (smaller ropes, and R2 restated to 4.5 kg if the ladder still exceeds 4.0 kg) and 9A (the 500 x 1,200 mm footing standard at every site). Four requirements also need trials with people before they can be closed (R2 throw accuracy, R5 climb time, R7 deploy time, R8 ageing); those are TRL 4 work. Status comes from the calculation note CNR-CAL-001; the tag in brackets is the line of `docs/04-calcs/sizing.py` that carries the figure.

Safety: CanalRung is safety-critical rescue and rope equipment. These requirements describe a research prototype, published as an open engineering reference and never as certified life-saving equipment. Meeting them on paper does not make the ladder safe to use; only proof loads, tank trials under lifeguard supervision and field trials can show that. Never enter the water to rescue someone: throw, anchor and call the emergency number.

■ Design envelope

The first version is sized for lined canals where the water is up to 2.0 m below the coping on a 1.5:1 (horizontal to vertical) side slope, or up to 3.0 m below the coping on a vertical wall, with surface currents up to 2.0 m/s and water from 5 to 30 °C. The design case in the model is the 1.5:1 slope with the water 2.0 m below the coping. Deeper slopes need a longer ladder (a later variant, CNR-DDR-001, D1).

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Verification	Status at TRL 3
R1	Reach the water from the bank	Rung section at least 4.0 m; at least two rungs at or below the water line on the design walls (1.5:1 slope to 2.0 m, vertical wall to 3.0 m)	Calculation [A1] to [A5]; deploy on a test wall at TRL 4	Met on paper: 4,020 mm; two rungs in the water on the design slope and four on a 3.0 m vertical wall; works on 1.5:1 slopes to 2.1 m and vertical walls to 3.8 m
R2	Throwable by one adult	Ladder as thrown (everything that leaves the box) at most 4.5 kg (9.9 lb), restated by Amish on 2026-10-03 from 4.0 kg; 8 of 10 untrained adults land the throw weight within 1 m of a target 3 m upstream	Weigh; throw trials with volunteers	Met on paper on mass: 4.27 kg [B3], 0.23 kg inside; throw accuracy needs trials
R3	Rungs float at the water line	Every rung floats with at least 0.15 kg to spare, with its share of rope; the bottom three rungs carry the throw weight with a factor of at least 1.5	Calculation [C1] to [C3]; tank test	Met on paper: 0.27 kg (plain) and 0.22 kg (stand-off) per rung; factor 1.73
R4	Carry a climbing adult	Each rung takes 1.5 kN at mid-span at no more than two thirds of yield; the two side ropes with their knots, wet, hold at least four times 4.5 kN (18 kN)	Calculation [D1], [E1]; proof-load test (TRL 4)	Met on paper: rung at 0.56 of yield; 9 mm ropes 18.4 kN, 4.08 times 4.5 kN
R5	Climbable on a flat wall	Clear width at least 280 mm; rung pitch 300 to 350 mm; at least 60 mm of hand room behind the rung at the stand-offs; an adult volunteer climbs out of a 1.5 m deep test tank with a smooth vertical wall in under 60 s	Geometry [A6]; supervised tank trials with lifeguards	Geometry met by design (298 mm, 335 mm, 70 mm); climb time needs trials
R6	Anchor holds	Station anchor (eye nut, bolt, post and the standard 500 x 1,200 mm footing used at every site) holds 4.5 kN with a factor of at least 1.5 on each element, in drained or saturated granular bank soil (restated by Amish on 2026-10-03)	Calculation [G1] to [G6]; pull test on site	Met on paper: eye nut 1.53; footing 3.58 drained, 1.59 saturated
R7	Deploy quickly	Lid open to ladder in the water in under 30 s, using only the pictogram on the lid	Timed trials with untrained volunteers	Not verifiable at TRL 3; estimate 15 s [H1]
R8	Survive outdoor storage	Ropes and rungs keep at least 80 % of strength after the equivalent of 2 years of outdoor exposure	Accelerated UV ageing and pull test	Met by design (stored in an opaque UV-stabilised box) [H2]; to confirm by test
R9	Value for money	Parts cost reported against the value-engineering target of USD 500, including the post, footing and box	Costed bill of materials [I1]	Value-engineering target: USD 500. Estimated cost of the constructable design: USD 605.20 (USD 105.20 over the target)
R10	Nothing thrown can hurt the	Soft throw weight; no exposed sharp edges or tube ends; a blunt-tip knife at the station to cut rope if anyone is caught	Inspection	Met by design [H3]

ID	Requirement	Target	Verification	Status at TRL 3
	person in the water			

■ Requirements not met or at risk

- **R2 (restated, now met on mass):** with 9 mm ropes the ladder as thrown weighs 4.27 kg, still 0.27 kg over the earlier 4.0 kg, so the target was restated to 4.5 kg under Amish's fallback in decision 4A (CNR-DDR-003). The stronger rungs were kept. Whether 4.3 kg is easy to throw is a question for the TRL 4 throw trials.
- **R4 (thin margin on the ropes):** the 9 mm EN 1891 type B ropes hold 18.4 kN wet with knots, 4.08 times 4.5 kN, just over the four times R4 asks for and above Amish's floor of three times. The bought rope must declare at least 18 kN.
- **R6 (soil outside the screening values):** the standard footing holds with factors of 3.58 in drained and 1.59 in saturated granular soil. Soft clay or peat banks need a site pull test.
- **R9 (over the target):** the larger standard footing adds USD 65.00 of concrete; the estimate is USD 105.20 over the value-engineering target.

■ Assumptions

- Many canal falls happen near places where a station can be placed: crossings, pump houses, paths and farms.
- A person in cold water can grip a rung long enough to climb if the rung is within reach.
- Throwing upstream and letting the current carry the ladder is easier for a bystander than throwing to the person.
- Irrigation districts would allow stations on their banks.
- The design climber is 100 kg with a dynamic factor of 1.5 (1.47 kN); the 4.5 kN anchor target is about three times that.